MATH317 · NUMERICAL METHODS

Chapter 3.1 Bisection Method

Midterm summary · Lecture 5

Muharib

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01 What is a root?

A **Root** (or **Zero**) of a function `f` is a number `r` with `f(r) = 0`.

Some equations can be solved by hand. Example:

$f(x) = 6x^2 - 7x + 2 = (2x - 1)(3x - 2)$ roots: $\frac{1}{2}$ and $\frac{2}{3}$

Most equations have no hand solution, for example $x^3 - 2\sin x = 0$. For those we need a numerical method.

Methods of the chapter

Chapter 3 has three methods: **Bisection**, **Newton** and **Secant**. This file covers bisection, together with **False Position**.

02 The theorem behind bisection

The method rests on the **Intermediate Value Theorem**:

Theorem

If f is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then there is at least one root between a and b.

In words: if the function has opposite signs at the two ends, it must cross the x-axis between them.

Watch out

If $f(a) \cdot f(b) > 0$ there may or may not be a root. A sign change is sufficient, not necessary.

03 The bisection algorithm

The idea: halve the interval each time and keep the half that contains the root. It is also called the **Binary-search method**.

1

Start with `[a, b]` and compute `u = f(a)` and `v = f(b)`. We need `u · v < 0`.

2 **Compute the midpoint $c = \frac{a + b}{2}$ and `w = f(c)`.**

3 **Check the sign of `u · w`:**

Case	Meaning	Update
`u · w < 0`	The root is in `[a, c]`	`b = c`
`u · w > 0`	The root is in `[c, b]`	`a = c`
`u · w = 0`	`c` is the root	stop

4 **Go back to step 2 and repeat.**

When to stop (stopping criteria)

- When the number of iterations exceeds `n_max`.
- Or when the **Error bound** drops below `ε`.

04 A full worked example

Slide example: $f(x) = x^3 - 3x + 1$ on $[0, 1]$.

$$f(0) = 1 \qquad f(1) = 1 - 3 + 1 = -1 \qquad f(0) \cdot f(1) < 0 \quad \rightarrow \quad \text{a root exists}$$

n	a	b	c	f(c)	Error
0	0	1	0.5	−0.375	0.5
1	0	0.5	0.25	0.2656	0.25
2	0.25	0.5	0.375	−0.0723	0.125
3	0.25	0.375	0.3125	0.0930	0.0625
4	0.3125	0.375	0.34375	0.0094	0.03125

How the first two rows are read:

- $n = 0$: the midpoint is 0.5 and $f(0.5) = -0.375$. Its sign is opposite to $f(0) = 1$, so the root is in $[0, 0.5]$.
- $n = 1$: the midpoint is 0.25 and $f(0.25) = 0.2656$. Its sign matches $f(0)$, so the root is in $[0.25, 0.5]$.

The true root is 0.3472963553. The method approaches it slowly.

The Error column

In the doctor's table the Error is half the length of the current interval: it starts at $\frac{b - a}{2}$ and halves on every row.

05 Error bound and number of steps

After n steps (counting from $n = 0$), the distance between the midpoint c_n and the root r satisfies:

$$| r - c_n | \leq \frac{b - a}{2^{n+1}}$$

How many steps for accuracy ε?

$$\frac{b - a}{2^{n+1}} < \varepsilon$$

$$n > \frac{\log(b - a) - \log(2\varepsilon)}{\log 2}$$

Slide example

How many steps give full single precision when $a = 16$ and $b = 17$?

$$16 = (10000.0)_2 \qquad 17 = (10001.0)_2$$

The root lies between them, so the first 5 bits are known. Single precision has 24 bits, leaving 19 to determine. The inequality gives:

$$n \geq 20$$

Appeared in the midterm

A statement in midterm 472 said the bisection error after n steps is at most $\frac{b - a}{2^n}$.

The answer was False; the correct power is 2^{n+1}.

Strengths and weaknesses of bisection

Strengths	Weaknesses
Reliable: it always closes in on a root once the sign condition holds.	Slow. It does not give the exact root in a finite number of steps.

That is why it is often used to get close to a root before switching to a faster method such as Newton's.

06 False Position (Regula Falsi)

Like bisection it keeps the root bracketed between `a` and `b`. The difference: instead of the midpoint, it uses the point where the line through `(a, f(a))` and `(b, f(b))` crosses the x-axis.

$$c = \frac{a \cdot f(b) - b \cdot f(a)}{f(b) - f(a)}$$

The new interval is chosen as in bisection:

Case	New interval
`f(a) · f(c) < 0`	`[a, c]`
`f(c) · f(b) < 0`	`[c, b]`

Slide example

`f(x) = x³ − 2x − 3` on `[0, 2]`.

$$f(0) = -3 \qquad f(2) = 1 \qquad f(0) \cdot f(2) = -3 < 0$$

$$\text{Step 1:} \quad c = \frac{0 \cdot 1 - 2 \cdot (-3)}{1 - (-3)} = \frac{6}{4} = 1.5$$

$$f(1.5) = -2.625 \quad \text{(negative)} \quad \rightarrow \quad a = 1.5\,, \quad b = 2$$

$$\text{Step 2:} \quad c = \frac{1.5 \cdot 1 - 2 \cdot (-2.625)}{1 + 2.625} = \frac{6.75}{3.625} = \frac{54}{29} = 1.8621$$

$$f(1.8621) = -0.2678 \quad \text{(negative)} \quad \rightarrow \quad a = 1.8621\,, \quad b = 2$$

07 How it shows up in the midterm

The bisection question appeared in all four midterms, worth 8 to 11 marks. Its shape is fixed: part (a) shows a root exists, part (b) applies the method with `n_max = 2`.

Midterm 472 solved the doctor's way

f(x) = $-x^2 + 4x - e^x$. (a) Prove that f has exactly one real root in [0, 1]. (b) Find a root on [0, 1] using the bisection method. Terminate if iterations exceed nmax = 2 or the error bound is less than 0.5×10^{-6}.

$$\text{(a)} \quad u = f(0) = -1 \qquad v = f(1) = 0.2817$$

$$u \cdot v < 0 \quad \rightarrow \quad f \text{ has a root in } [0, 1]$$

(b) $n = 0$: $x_0 = \frac{0 + 1}{2} = 0.5$ $\quad w = f(0.5) = 0.1013$

$u \cdot w < 0 \quad \rightarrow \quad b = x_0 \quad \rightarrow \quad [0, 0.5]$

$n = 1$: $x_1 = \frac{0 + 0.5}{2} = 0.25$ $\quad w = f(0.25) = -0.3465$

$u \cdot w > 0 \quad \rightarrow \quad a = x_1 \quad \rightarrow \quad [0.25, 0.5]$

$n = 2$: $x_2 = \frac{0.25 + 0.5}{2} = 0.375$ $\quad w = f(0.375) = -0.0956$

$u \cdot w > 0 \quad \rightarrow \quad a = x_2 \quad \rightarrow \quad [0.375, 0.5]$

n	x_n	$f(x_n)$	Error
0	0.5	0.1013	0.5
1	0.25	−0.3465	0.25
2	0.375	−0.0956	0.125

Notes on the answer key

The key prints `x₂ = 2.75` in the table and an error of `0.25` for the third step in the text. Both are typos; the correct values are `0.375` and `0.125`.

The doctor also notes that skipping any step loses marks. Write every line.

08 Common mistakes

Mistake	Fix
Calculator in degree mode with `sin` or `cos`	It must be in radian mode.
Writing the error bound as $\frac{b-a}{2^n}$	The correct bound is $\frac{b-a}{2^{n+1}}$.
Keeping the wrong half of the interval	Keep the half whose ends have opposite signs.
Writing only two rows when `n_max = 2`	Three rows: `n = 0, 1, 2`.
Saying bisection gives the exact root	It gives an approximation; the error halves each step.

09 Check yourself

1. `f(x) = x³ − x − 1` on `[1, 2]`. Apply bisection with `n_max = 2`.
2. `f(x) = cos x − x` on `[0, 1]` (radians). Apply bisection with `n_max = 2`.
3. How many steps on `[0, 1]` make the error smaller than `10⁻³`?

Solutions

Q1 · n	c	f(c)	Error	Next interval
0	1.5	0.875	0.5	[1, 1.5]
1	1.25	−0.2969	0.25	[1.25, 1.5]
2	1.375	0.2246	0.125	[1.25, 1.375]

Q2 · n	c	f(c)	Error	Next interval
0	0.5	0.3776	0.5	[0.5, 1]
1	0.75	−0.0183	0.25	[0.5, 0.75]
2	0.625	0.1860	0.125	[0.625, 0.75]

Q3) $\frac{1-0}{2^{n+1}} < 10^{-3} \rightarrow 2^{n+1} > 1000 \rightarrow n+1 \geq 10$

$n \geq 9$